

An Empirical Study of Regularization, Algorithmic Stability and Generalization in Ridge Regression

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Abstract

This project studies the relationship between regularization strength, algorithmic stability and generalization performance in supervised learning. I evaluated and implemented the ridge regression on both a synthetic dataset and a real world dataset. Algorithmic stability was estimated through a leave-one-out retraining procedure that measures the average change in predictions after removing individual training observations. The results show that moderate L2 regularization generally improves model stability and this leads to reduction in the test error . An additional analysis is the effect of the dataset size which illustrates how increasing the amount of training data improves stability. Overall, the findings provide empirical support for the theoretical framework proposed by Bousquet and Elisseeff (2002), demonstrating the close relationship between regularization, stability and generalization.

1 Introduction

A fundamental challenge in supervised learning is the gap between *Empirical Risk Minimization* (ERM), which seeks to minimize the prediction error on the observed training data, and *generalization*, which studies the model's ability to perform accurately on unseen data. Classical theory explains this relationship using concepts such as uniform convergence and complexity measures, Following this theory , we expect simpler models to generalise better.

However, these classical explanations become less informative in settings involving highly correlated features, high-dimensional data, or overparameterized models. In such situations, the theory may predict poor generalization even when learning algorithms generalise well empirically. This suggests that controlling model complexity alone is not sufficient to fully explain why certain algorithms generalize successfully.

An alternative perspective was proposed by Bousquet and Elisseeff (2002), who studied generalization through the concept of *algorithmic stability*. Rather than focusing on how complex the hypothesis class is, algorithmic stability measures how sensitive a learning algorithm is, to small perturbations of the training dataset. A learning algorithm is considered stable if removing or modifying a single training observation results in only a small change in the learned model and its predictions. According to this framework, more stable algorithms are expected to generalize better.

The objective of this project is to empirically investigate the relationship between regularization strength, algorithmic stability and generalization performance . Ordinary Least Squares

(OLS) is used as the unregularized baseline, while Ridge Regression introduces an L_2 regularization penalty that shrinks the regression coefficients and is expected to reduce the sensitivity of the model to individual training observations. The empirical analysis is first conducted on a synthetic regression dataset specifically designed to exhibit strong multicollinearity, allowing the theoretical effects of regularization to be studied under controlled conditions. The same experimental procedure is then applied to the Communities and Crime dataset to assess whether similar behaviour is observed in a realistic setting. As an additional extension, I study the effect of dataset size on stability.

The remainder of this report is organized as follows. Section 2 introduces the methodology, including the theoretical foundations of Ordinary Least Squares, Ridge Regression, algorithmic stability, and the experimental setup. Sections 3 and 4 present the empirical results obtained from the synthetic dataset and the Communities and Crime dataset, respectively. Section 5 investigates the effect of training set size on algorithmic stability and generalization. Section 6 discusses the results in the context of the theoretical framework proposed by Bousquet and Elisseeff (2002). Finally, Section 7 concludes the report by summarizing the main findings.

2 Methods

2.1 Models

Two linear regression models are considered in this study: Ordinary Least Squares (OLS) and Ridge Regression. OLS serves as the unregularized baseline, while Ridge Regression introduces an L_2 regularization term to investigate how regularization influences algorithmic stability and generalization performance. Both models were implemented from scratch, allowing full control over the estimation procedure.

2.1.1 Ordinary Least Squares (OLS)

Ordinary Least Squares (OLS) serves as the unregularized baseline model throughout this study. The objective of OLS is to estimate the regression coefficients that minimize the sum of squared residuals between the observed target values and the predictions produced by the linear model. Under the assumption of a linear relationship between the predictors and the response variable, the parameter vector can be obtained analytically through a closed-form solution.

Given a design matrix $S \in \mathbb{R}^{n \times d}$ containing n observations and d features, together with a target vector $y \in \mathbb{R}^n$, the OLS parameter vector w is estimated by solving the following optimization problem:

$$\min_w \|y - Sw\|^2,$$

By setting the gradient of this objective function with respect to w to zero, we obtain the standard closed-form normal equation

$$w = (S^T S)^{-1} S^T y.$$

Here, $S^T S$ is commonly referred to as the Gram matrix and summarizes the relationships between the predictor variables. When this matrix is invertible, the solution provides the set of coefficients that minimizes the residual sum of squares.

In practice, however, the matrix $S^T S$ may be singular or nearly singular, particularly when the predictors are highly correlated or when the dataset contains redundant features. Computing the standard matrix inverse in these situations can lead to numerical instability and unreliable parameter estimates. To improve numerical robustness, the Moore–Penrose pseudo-inverse is used instead of the ordinary matrix inverse. In the implementation, this is computed using the NumPy function `np.linalg.pinv`, which provides a stable solution even when the Gram matrix is ill-conditioned.

The implementation follows three main steps. First, the Gram matrix $S^T S$ is computed from the design matrix. Second, the Moore–Penrose pseudo-inverse of this matrix is calculated. Finally, the resulting matrix is multiplied by $S^T y$ to obtain the regression coefficient vector.

Once the coefficients have been estimated, predictions are obtained through

$$\hat{y} = Sw,$$

where $\hat{y}$ denotes the predicted response vector.

2.1.2 Ridge Regression

While OLS minimizes only the residual error, it can become highly sensitive to small perturbations of the training data when the predictors are strongly correlated or when the Gram matrix is nearly singular. Ridge Regression addresses this limitation by introducing an L_2 regularization term that penalizes large regression coefficients. By shrinking the coefficients towards zero, Ridge Regression improves the conditioning of the optimization problem, leading to a more stable estimator with lower variance.

The Ridge objective function is

$$\min_w (\|y - Sw\|^2 + \alpha \|w\|^2),$$

where $\alpha \geq 0$ is the regularization parameter controlling the strength of the penalty. The corresponding closed-form solution is

$$w = (S^T S + \alpha I)^{-1} S^T y,$$

where I denotes the identity matrix.

The addition of the regularization term increases the diagonal elements of the Gram matrix, making it better conditioned and less sensitive to multicollinearity. As a result, small changes in the training data produce smaller changes in the estimated regression coefficients, which is expected to improve algorithmic stability and generalization performance.

The implementation closely follows that of OLS. First, the Gram matrix $S^T S$ is computed. The regularization matrix αI is then added before computing the Moore–Penrose pseudo-inverse using `np.linalg.pinv`. Finally, the resulting matrix is multiplied by $S^T y$ to estimate the regression coefficients.

The intercept term is excluded from regularization because it represents the baseline level of the response rather than the contribution of a predictor. This is implemented by modifying the identity matrix and setting its first diagonal element to zero before applying the regularization penalty.

Predictions are then obtained using

$$\hat{y} = Sw,$$

in the same way as for Ordinary Least Squares.

2.2 Stability Measure

Stability is estimated empirically using a leave-one-out retraining procedure. For each value of the regularization parameter, the model is first trained using the complete training set and predictions are computed on a fixed test set. Each training observation is then removed one at a time, the model is retrained, and a new set of predictions is obtained using the same test data. The difference between the original and retrained predictions quantifies the influence of the removed observation on the learned model.

The empirical stability metric is defined as

$$\text{Stability} = \frac{1}{n} \sum_{i=1}^n \frac{1}{m} \sum_{j=1}^m \left| \hat{y}_j^{(-i)} - \hat{y}_j \right|,$$

where n denotes the number of training observations, m is the number of test observations, $\hat{y}_j$ represents the prediction obtained using the full training set, and $\hat{y}_j^{(-i)}$ denotes the prediction obtained after removing the i -th training observation.

Lower values of this metric indicate greater algorithmic stability because the predictions change only slightly when the training data are perturbed. Conversely, larger values indicate that the learned model is more sensitive to individual observations. Since stable algorithms are expected to produce more consistent predictions on unseen data, this measure provides an empirical way of studying the relationship between regularization, stability, and generalization in both the synthetic and real-world experiments.

2.3 Experimental Setup

The same experimental procedure is applied to both the synthetic dataset and the Communities and Crime dataset in order to make the results comparable across settings. Each dataset is split into a training set and a test set using an 80/20 split. The training set is used to estimate the model parameters, while the test set is kept fixed and used only to evaluate generalization performance and compute the stability measure.

Before training the models, all predictor variables are standardized. The mean and standard deviation are computed using only the training set, and the same transformation is then applied to the test set. This avoids information leakage from the test set and ensures that all features are placed on a comparable scale.

After standardization, an intercept column is added to the design matrix. The intercept is included in both OLS and Ridge Regression, but it is excluded from the Ridge penalty because it represents the baseline level of the response variable rather than the effect of a predictor.

For each dataset, OLS is estimated as the baseline model. Ridge Regression is then estimated for a sequence of regularization parameters,

$$\alpha \in \{0, 0.001, 0.01, 0.1, 1, 10, 50, 100\}.$$

The case $\alpha = 0$ corresponds to the unregularized OLS solution, while larger values of α impose stronger regularization. This grid allows the analysis to compare the effect of weak, moderate and strong regularization on both prediction error and stability.

Model performance is evaluated using Mean Squared Error (MSE), defined as

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2.$$

Both training MSE and test MSE are reported. The training MSE measures how well the model fits the observed training data, while the test MSE measures generalization performance on unseen data.

For every value of α , the empirical stability metric described in Section 2.2 is computed using the leave-one-out retraining procedure. The relationship between regularization, stability and generalization is then analysed by comparing stability values with the corresponding test errors.

3 Controlled Experiment: Synthetic Dataset

3.1 Dataset Construction

Before evaluating the proposed methodology on a real-world dataset, a controlled synthetic dataset is used to isolate the effect of regularization on algorithmic stability and generalization. Synthetic data provide complete control over the statistical properties of the predictors, making it possible to deliberately construct situations in which Ordinary Least Squares is expected to perform poorly and Ridge Regression is expected to offer clear advantages.

The dataset consists of 200 observations and 80 predictor variables. The first 40 features are independently sampled from a standard normal distribution,

$$X_j \sim \mathcal{N}(0, 1),$$

while the remaining 40 features are generated as noisy copies of the first 40,

$$X_{j+40} = X_j + \varepsilon,$$

where ε is Gaussian noise with small variance.

This construction deliberately introduces strong multicollinearity among the predictor variables. Since many features contain almost identical information, the Gram matrix $S^T S$ becomes

poorly conditioned. Under these circumstances, the OLS estimator is expected to be highly sensitive to small perturbations of the training data, providing an ideal setting for studying the effect of regularization .

The target variable is generated according to

$$y = Xw + \epsilon,$$

where w is a randomly generated coefficient vector and ϵ is additive Gaussian noise. The inclusion of noise produces a more realistic regression problem by preventing the relationship between the predictors and the response from being perfectly deterministic.

The dataset is randomly divided into training and test sets using an 80/20 split. The training set is used to estimate the model parameters, while the test set remains fixed throughout all experiments. Keeping the test set fixed is particularly important because the stability measure compares predictions obtained from models trained on slightly different versions of the same training set. Using the same evaluation data ensures that any observed differences in the predictions are entirely attributable to changes in the training set.

3.2 Results

Table 1 summarizes the training Mean Squared Error (MSE), test MSE and empirical stability obtained for OLS and Ridge Regression across different values of the regularization parameter α .

Table 1: Synthetic dataset results

α	Model	Train MSE	Test MSE	Stability
0	OLS	0.4490	1.8987	0.0696
0.001	Ridge	0.4567	1.8166	0.0535
0.01	Ridge	0.5279	1.6810	0.0351
0.1	Ridge	0.6104	1.6392	0.0314
1	Ridge	0.6293	1.6110	0.0310
10	Ridge	0.7203	1.5342	0.0317
50	Ridge	2.1824	3.3525	0.0456
100	Ridge	4.9898	7.3671	0.0569

The results show a clear trade-off between model fit, stability and generalization. As the regularization strength increases, the training error gradually rises because the regression coefficients are increasingly constrained. In contrast, the test error initially decreases, reaching its minimum for an intermediate value of the regularization parameter before increasing again for large values of α . The empirical stability metric follows a similar pattern: moderate regularization substantially reduces the sensitivity of the model to individual training observations, whereas excessive regularization begins to deteriorate predictive performance. These findings are consistent with the theoretical expectation that an appropriate amount of regularization can improve generalization by increasing algorithmic stability without introducing excessive bias.

The behaviour observed in Table 1 is further illustrated by the following figures.

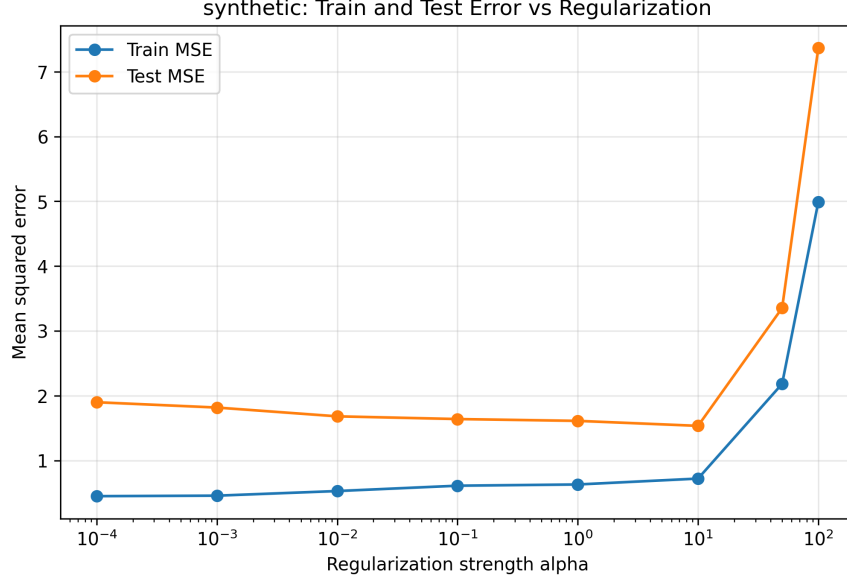


Figure 1: Training and test error versus regularization strength on the synthetic dataset.

Figure 1 shows that moderate regularization reduces the test error relative to OLS. The lowest test error is achieved around $\alpha = 10$. As expected, the training error increases monotonically with the regularization strength because larger values of α impose stronger constraints on the regression coefficients. For larger values of α , particularly $\alpha = 50$ and $\alpha = 100$, the test error increases sharply, indicating that the model becomes overly constrained and begins to underfit the data.

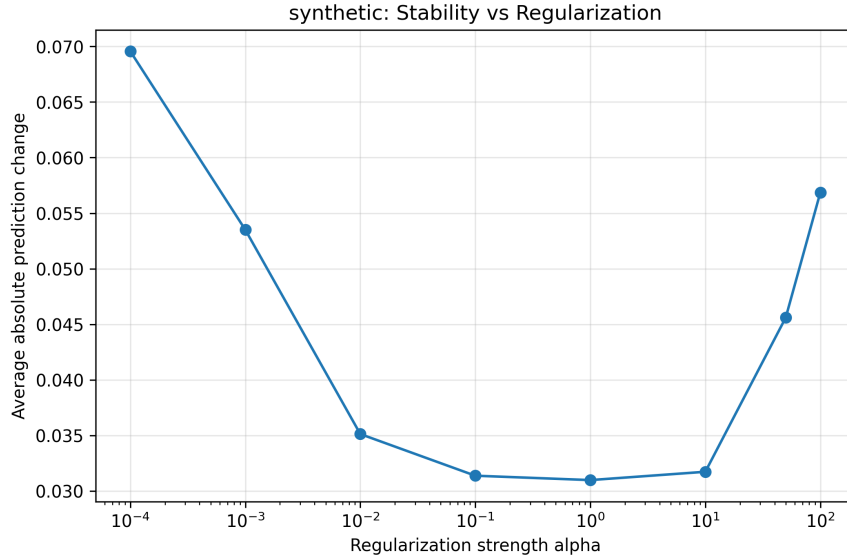


Figure 2: Stability versus regularization strength on the synthetic dataset.

Figure 2 shows that algorithmic stability improves substantially when moving from OLS to moderate levels of Ridge regularization. The stability metric decreases from 0.0696 for OLS to approximately 0.031 for $\alpha = 1$, indicating that the model becomes considerably less sensitive to the removal of individual training observations. For very large values of α , however, the

stability metric begins to increase again, suggesting that excessive regularization can distort the fitted model and negatively affect predictive behaviour.

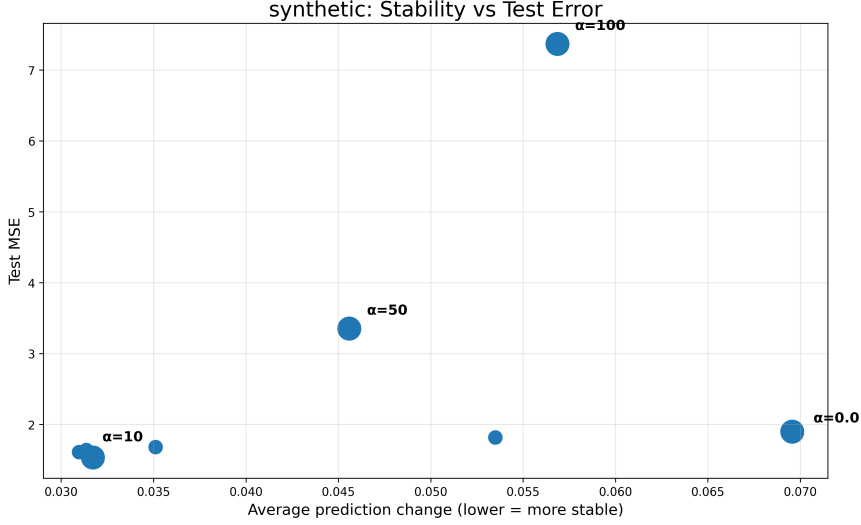


Figure 3: Stability versus test error on the synthetic dataset.

Figure 3 shows the relationship between algorithmic stability and generalization. The lowest test error is achieved at around $\alpha = 10$, This is the point that shows the optimum bias–variance tradeoff. The figure clearly highlights that maximizing stability alone is not sufficient: excessive regularization introduces additional bias, leading to underfitting and a deterioration in predictive performance.

4 Real-World Validation: Communities and Crime Dataset

4.1 Dataset Description

While the synthetic experiment provides a controlled environment for studying the relationship between regularization, algorithmic stability and generalization, it does not capture the complexity of real-world data. To evaluate whether the observed behaviour extends beyond an artificial setting, the same experimental procedure is applied to the Communities and Crime dataset obtained from the UCI Machine Learning Repository.

The dataset contains socio-economic, demographic and law-enforcement variables collected from communities across the United States. The response variable is the normalized violent crime rate, while the predictors describe characteristics such as population demographics, income, education, employment, housing, family structure and public safety. Since many of these variables measure related socio-economic phenomena, substantial correlations are expected among the predictors.

This dataset is particularly well suited to the objectives of this study because multicollinearity is common in socio-economic data. Variables describing poverty, unemployment, income, education and housing often contain overlapping information, making the design matrix potentially ill-conditioned. Consequently, Ordinary Least Squares is expected to be more sensitive

to small perturbations of the training data, while Ridge Regression should improve numerical stability through coefficient regularization.

Several preprocessing steps were applied before fitting the models. Non-predictive identifier variables were removed, and all variables containing more than 50% missing observations were discarded. The remaining missing values were imputed using the mean of the corresponding feature. After preprocessing, the dataset contains 1,994 observations and approximately 100 predictor variables.

4.2 Results

Table 2 summarizes the training Mean Squared Error (MSE), test MSE and empirical stability obtained for OLS and Ridge Regression across different values of the regularization parameter.

Table 2: Communities and Crime dataset results

α	Model	Train MSE	Test MSE	Stability
0	OLS	0.016196	0.019562	0.000457
0.001	Ridge	0.016196	0.019562	0.000457
0.01	Ridge	0.016196	0.019561	0.000457
0.1	Ridge	0.016197	0.019548	0.000454
1	Ridge	0.016210	0.019468	0.000438
10	Ridge	0.016347	0.019319	0.000390
50	Ridge	0.016675	0.019351	0.000326
100	Ridge	0.016917	0.019444	0.000292

Compared with the synthetic experiment, the effect of regularization is more gradual but follows the same overall pattern. The training error increases slightly as the regularization strength grows, reflecting the additional bias introduced by the Ridge penalty. At the same time, the test error decreases from the OLS solution to moderate values of α , reaching its minimum around $\alpha = 10$, before increasing slightly for stronger regularization. The empirical stability metric decreases consistently as α increases, indicating that Ridge Regression makes the model progressively less sensitive to small perturbations of the training data. Although the numerical improvements are smaller than those observed in the synthetic dataset, they provide evidence that the theoretical benefits of regularization also extend to a realistic regression problem.

The behaviour observed in Table 2 is further illustrated by the following figures.

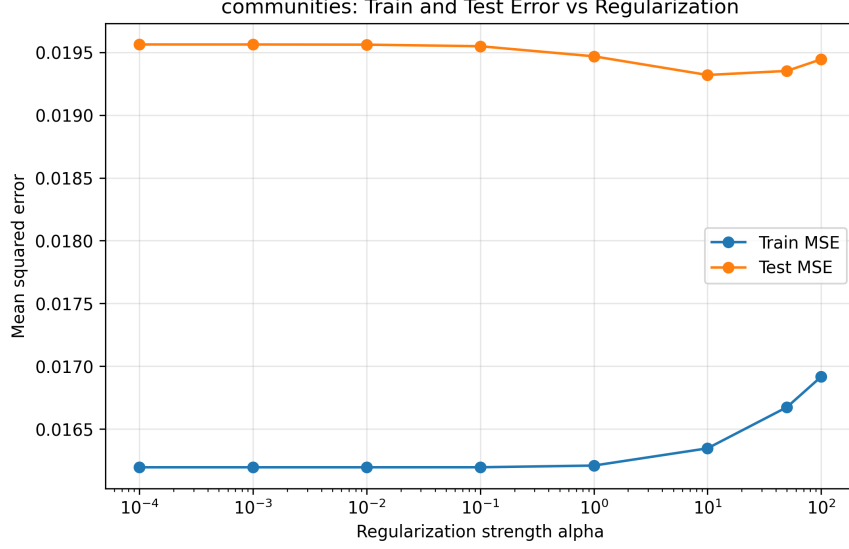


Figure 4: Training and test error versus regularization strength on the Communities and Crime dataset.

Figure 4 shows that the training error increases gradually as the regularization strength increases, while the test error initially decreases before reaching its minimum around $\alpha = 10$. Unlike the synthetic experiment, the improvement in predictive performance is relatively modest, reflecting the greater complexity and noise present in real-world data. Nevertheless, the results indicate that a moderate amount of regularization is sufficient to improve generalization without substantially increasing the bias of the model.

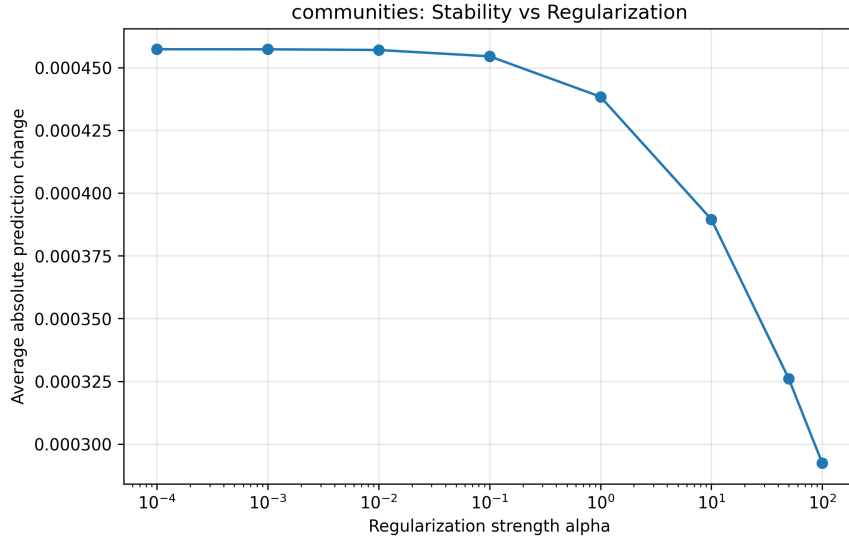


Figure 5: Stability versus regularization strength on the Communities and Crime dataset.

Figure 5 demonstrates that algorithmic stability improves steadily as the regularization parameter increases. Unlike the synthetic dataset, where the stability metric eventually rises again for very large values of α , the empirical stability continues to decrease throughout the range of regularization strengths considered. This suggests that, for this dataset, stronger regularization consistently reduces the influence of individual training observations on the fitted

model. The smoother behaviour is likely due to the larger sample size and the more complex correlation structure of the predictor variables.

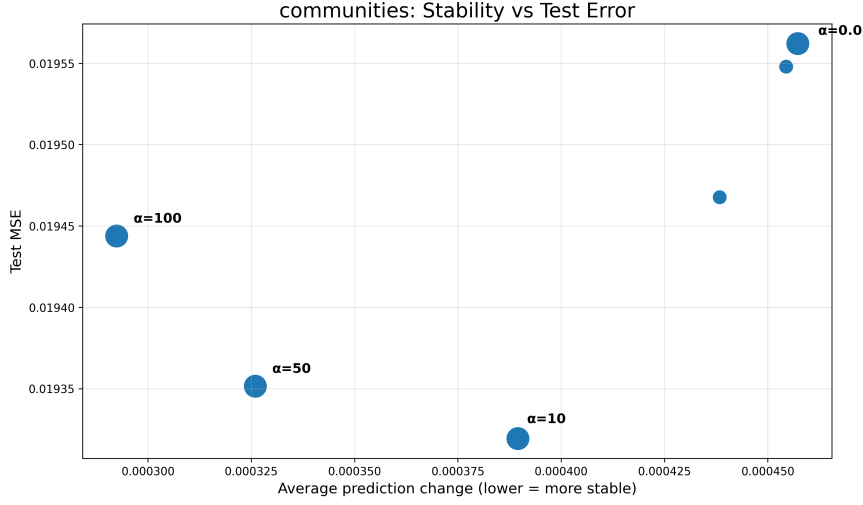


Figure 6: Stability versus test error on the Communities and Crime dataset.

Figure 6 illustrates the relationship between algorithmic stability and predictive performance on the real-world dataset. As regularization improves stability, the test error decreases until an optimal level of α is reached. Beyond this point, additional regularization produces only marginal gains in stability while slightly increasing the prediction error. These findings reinforce the conclusions obtained from the synthetic experiment: improving algorithmic stability generally leads to better generalization, although excessive regularization may eventually introduce bias to offset these benefits. “

5 Extension: Effect of Training Set Size

The previous experiments investigated how regularization affects algorithmic stability while keeping the amount of training data fixed. This additional experiment studies a complementary question: how does the size of the training set influence stability and generalization?

From the perspective of algorithmic stability, the influence of an individual training observation is expected to decrease as the dataset becomes larger. When only a small number of observations are available, removing a single training point represents a relatively large perturbation of the dataset and may substantially affect the learned model. As the number of training observations increases, the contribution of each individual sample becomes smaller, leading to a more stable estimator.

To investigate this effect, Ridge Regression is trained using different subsets of the Communities and Crime dataset while keeping the regularization parameter fixed at $\alpha = 1$. Training set sizes of 100, 300, 600, 1000 and the full training set (1595 observations) are considered. For each training size, the model is trained using the same experimental procedure described in Section 3, and the training error, test error and empirical stability are computed.

5.1 Results

Table 3 reports the training Mean Squared Error (MSE), test MSE and empirical stability obtained for different training set sizes.

Table 3: Effect of training set size on model performance and stability

Training Size	Train MSE	Test MSE	Stability
100	0.00439	0.03540	0.01337
300	0.00957	0.02927	0.00284
600	0.01264	0.02134	0.00127
1000	0.01566	0.01976	0.00072
1595	0.01621	0.01947	0.00044

The results demonstrate a clear relationship between the amount of training data, predictive performance and algorithmic stability. As the training set becomes larger, the empirical stability metric decreases substantially, indicating that the model becomes progressively less sensitive to the removal of individual observations. At the same time, the test error consistently decreases, suggesting that additional training data improve the model’s ability to generalize to unseen observations. Although the training error increases slightly as the dataset grows, this is expected because larger training sets capture a broader range of variability and are therefore more difficult to fit perfectly.

The trend observed in Table 3 is illustrated in the figure below.

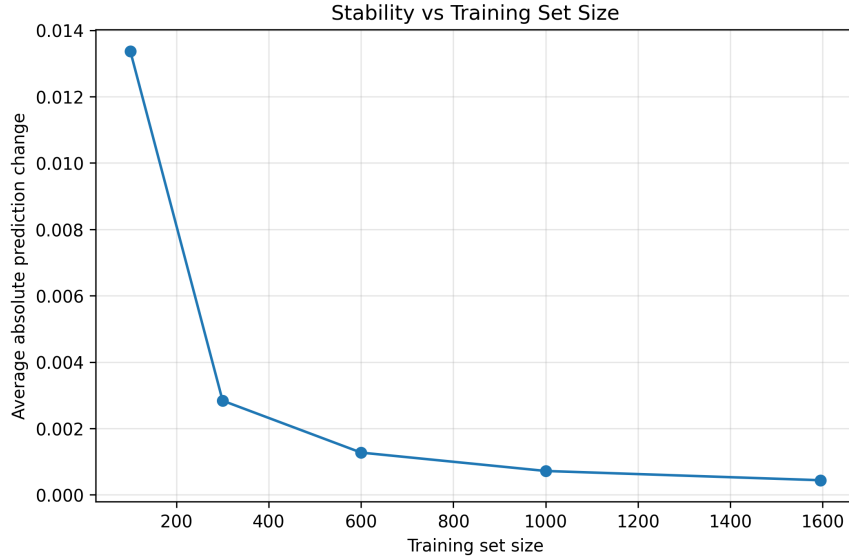


Figure 7: Empirical stability as a function of training set size.

Figure 7 shows that algorithmic stability improves rapidly as additional training observations are included. The largest improvement occurs when increasing the training size from 100 to approximately 600 observations, after which the stability measure decreases more gradually. This behaviour suggests diminishing returns: while adding more data continues to improve stability, the marginal benefit becomes smaller as the dataset grows. The results are consis-

tent with the theoretical expectation that the influence of any individual training observation decreases as the sample size increases.

6 Discussion

The objective of this study was to investigate how regularization influences algorithmic stability and generalization in linear regression. Across both the synthetic and real-world experiments, the results consistently showed that introducing L_2 regularization through Ridge Regression improves the stability of the learning algorithm while also reducing the prediction error compared to the unregularised least squares .

The synthetic dataset provided a controlled environment in which the effects of regularization were particularly evident. By introducing strong multicollinearity, the experiment highlighted the instability of OLS and demonstrated how Ridge Regression improves the conditioning of the optimization problem. The same qualitative behaviour was observed on the Communities and Crime dataset. Although the improvements were smaller than those obtained on the synthetic data, moderate regularization again produced the lowest test error together with improved algorithmic stability. Furthermore, the dataset-size extension showed that increasing the amount of training data further improves stability and generalization by reducing the influence of individual observations on the fitted model.

The experiments also revealed several important observations regarding the limitations of regularization. While moderate values of the regularization parameter consistently improved both stability and predictive performance, excessively large values of α increased the test error. This behaviour illustrates the classical bias–variance trade-off: reducing variance through regularization is beneficial only up to a certain point, after which the increase in bias leads to underfitting. In addition, the improvements observed on the Communities and Crime dataset were considerably smaller than those obtained on the synthetic dataset, indicating that regularization addresses only one source of prediction error. Real-world datasets contain additional challenges, such as measurement noise, missing values and more complex relationships between variables, which cannot be completely overcome through coefficient regularization alone.

References

- [1] Bousquet, O. and Elisseeff, A. (2002). *Stability and Generalization*. Journal of Machine Learning Research, 2, 499–526.
- [2] UCI Machine Learning Repository. *Communities and Crime Data Set*. Available at: <https://archive.ics.uci.edu/dataset/183/communities+and+crime>

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